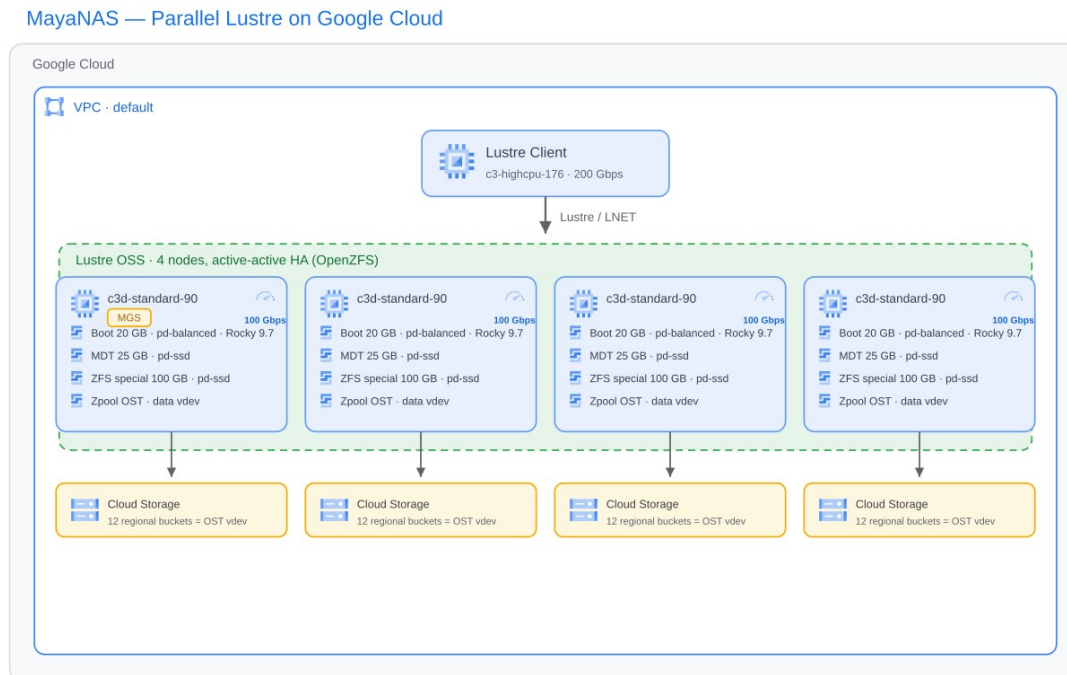


MayaNAS Lustre — MLPerf Storage v3.0 (UNet3D)

Submitter: ZettaLane Systems · **Division:** Closed · **System:** mayanas-lustre-gcp_c3hc176_c3d90_1c_4s

architecture



MayaNAS Storage Cluster (4 nodes, active-active HA)

- 4 × GCP c3d-standard-90 (AMD EPYC 9B14 Turin, 90 vCPU / 45 cores, 354 GB RAM)
- Rocky Linux 9.7 · 100 Gbps Tier_1 networking
- Each node: Lustre OSS with 1 OST (OpenZFS zpool; data vdev = 12 Standard regional GCS buckets via objbacker.io)
- Per node: Boot 20 GB pd-balanced · MDT 25 GB pd-ssd · ZFS special 100 GB pd-ssd
- MGS on node 1's VIP · regional buckets enable HA failover (surviving node imports partner zpool)

Benchmark Client

- 1 × GCP c3-highcpu-176 (Intel Xeon Platinum 8481C, 176 vCPU, 346 GB RAM)
- Ubuntu 24.04 · 200 Gbps networking
- Lustre client, O_DIRECT reads, read_threads=128

Results

- UNet3D (na=3, 3× B200): mean Accelerator Utilization ~92%

Availability

MayaNAS and MayaScale can be deployed from the Google Cloud Marketplace portal or as Terraform modules (<https://github.com/zettalane-systems/open-lustre-cloud>). Contact: info@zettalane.com.